MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 4: Continuity. Analytic functions.

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Lecture outline

1. Continuity
2. Uniform Continuity
3. Differentiable and continuous functions
4. Analytic functions

Continuity: preliminaries

Let M be an arbitrary point set in the complex plane. Suppose that z is allowed to denote any point of M. Then z is called a complex variable in M and M is called the region of variation of z. (If $z = x + iy$, then x and y are said to be *real* variables of S.)

If $\exists$ a rule by means of which a new number w may be associated with each point z of M, w is called a *function* of the complex variable z and we write

$$w = f(z)$$

where "f" stands for the prescribed rule. We say $f(z)$ *is defined in* M. M is called the *region of definition* of the function; z is called the *argument* of the function; the totality of values w which correspond to the points z of M is called the *range of values* of the function (over M).

For the purposes of defining continuity and differentiability we usually specialize M to be a domain D (open, connected set).

Continuity: preliminaries 2

We may have *many-valued* functions (more than one number w is associated with each $z \in M$) or *single-valued* functions (a unique number w is associated with each $z \in M$.)

Example: For a positive integer $n \geq 2$, $f(z) = z^{\frac{1}{n}}$ is an $n-$valued function for $z \neq 0$. But

$$f(z) = |z|^{\frac{1}{n}} (\cos \frac{\theta}{n} + i \sin \frac{\theta}{n}) \qquad (0 \leq \theta < 2\pi)$$

is a single valued function of z for all finite z; indeed, even in the extended $z - plane$ if we set $f(\infty) = \infty$.

In general we assume that if $f(z)$ is defined in a domain D that this implies $f(z)$ is single-valued.

We write $z = x + iy$ and $w = u + iv$. Then the relation $w = f(z)$ may also be interpreted as follows: To the pair of real numbers x, y there corresponds, by certain rules, two new real numbers u, v. Thus u and v are defined as a pair of real functions of two real variables, x and y. We set $u = u(x, y)$ and $v = v(x, y)$. Hence

$$f(z) = u(x, y) + iv(x, y)$$

Continuity: limit-based definition

Definition

We say that a complex *single-valued* function $f(z)$ defined in a domain D *approaches the limit* A *as* z *approaches a point* z_0 in D and we write

$$\lim_{z \to z_0} f(z) = A$$

or $f(z) \to A$ as $z \to z_0$ if given any $\epsilon > 0$ $\exists$ a number $\delta = \delta(\epsilon) > 0$ such that

$$|f(z) - A| < \epsilon$$

for all z such that

$$0 < |z - z_0| < \delta.$$

Suppose that in addition $A = f(z_0)$, so that

$$\lim_{z \to z_0} f(z) = f(z_0).$$

Then we say that $f(z)$ is *continuous at* z_0.

Continuity: comments to the definition 1

Remark: Thus $f(z)$ is said to be *continuous* at z_0 if given $\epsilon > 0$ $\exists$ a number $\delta(\epsilon, z_0) > 0$ such that

$$|f(z) - f(z_0)| < \epsilon$$

$\forall$ z such that

$$|z - z_0| < \delta$$

Geometrically this means that the values of $w = f(z)$ at every point of a sufficiently small neighborhood $|z - z_0| < \delta$ of the point z_0 all lie inside an arbitrarily small neighborhood $|w - w_0| < \epsilon$ of the point $w_0 = f(z_0)$.

Continuity: comments to the definition 2

Remarks: Since the definition of continuity for complex functions is formally the same as for real functions, the proofs of the familiar theorems concerning algebraic operations on continuous functions carry over from the real case to the complex case. Thus if $f(z)$ and $g(z)$ are both continuous at a point z_0, then so is $f(z) \pm g(z)$, $f(z)g(z)$, and $f(z)/g(z)$ provided that $g(z_0) \neq 0$ in the quotient. Moreover, if the function $f(z)$ is continuous at the point z_0 and if the function $\varphi(w)$ is continuous at the point $w_0 = f(z_0)$, then the composite function $\varphi(f(z))$ is continuous at the point z_0.

Continuity: example

Example

The function $\quad w = z^n \quad (n = 1, 2, 3, \ldots)$
is continuous in the whole finite plane. In fact, setting $w_0 = z_0^n$ where z_0 is any finite point, we have

$$w - w_0 = z^n - z_0^n = (z - z_0)(z^{n-1} + z^{n-2}z_0 + \cdots + z_0^{n-1}).$$

$\Longrightarrow \quad |w - w_0| = |z - z_0| \left|z^{n-1} + z^{n-2}z_0 + \cdots + z_0^{n-1}\right| \leq$
$|z - z_0| \left(r^{n-1} + r^{n-2}r_0 + \cdots + r_0^{n-1}\right)$
where $r = |z|$ and $r_0 = |z_0|$. But $|z - z_0| < \delta \Longrightarrow$

$$r = |z| = |z_0 + (z - z_0)| \leq |z_0| + |z - z_0| < r_0 + \delta$$

Hence
$|w - w_0| < |z - z_0| \left[(r_0 + \delta)^{n-1} + (r_0 + \delta)^{n-2}r_0 + \cdots + r_0^{n-1}\right] < n\delta(r_0 + \delta)^{n-1}.$
It follows that $|w - w_0|$ can be made less than any preassigned positive number $\in$ by choosing δ suitably. Hence the function $w = z^n$ is continuous at every point $z_0 \neq \infty$.

Continuity on the domain

Definition

A complex function $f(z)$ is said to be continuous in the domain D if it is continuous at every point of D.

Remarks: When a complex function $f(z)$ is also defined at some boundary points of D,we speak of "continuity from the interior".
$f(z) = u(x, y) + iv(x, y)$ is continuous $\iff$ $u(x, y)$ and $v(x, y)$ are continuous functions of the real variables x and y.

Uniform continuity : definition

In general in the definition of continuity $\delta = \delta(\epsilon, z_0)$, that is, δ depends both upon the given value of ϵ and the point z_0 under consideration. However, in certain cases δ depends *only* upon ϵ. This situation is called *uniform continuity.*

Definition

A complex function $f(z)$ defined in a domain D is said to be *uniformly continuous* in D if given $\in > 0$ $\exists$ a number $\delta = \delta(\in)$ such that

$$|f(z') - f(z'')| < \in$$

for *all* points z', z'' in D such that

$$|z' - z''| < \delta$$

Uniform continuity in a closed bounded domain

Remark: Unlike the case of ordinary continuity, it is meaningless to talk about uniform continuity at a single point z_0. The key observation here is that the δ in the above definition must be *independent* of the points z' and z'' in D. If $f(z)$ is uniformly continuous in D, then it is certainly continuous in D. However, a function can be continuous in D without being uniformly continuous in D. In the real case continuity in a closed bounded domain implies uniform continuity there. We have a similar result in the complex case.

Theorem

If $f(z)$ is continuous in a closed bounded domain $\bar{D}$, then it is uniformly continuous in $\bar{D}$.

Differentiability and continuity

Definition

A function $w = f(z)$ defined in a domain D is said to be *differentiable* at a point $z \in D$ if

$$\lim_{h \to 0} \frac{f(z+h) - f(z)}{h} \qquad (1)$$

exits and its value is *independent* of the way in which $h \to 0$. This limit is denoted by $f'(z)$ or $\frac{dw}{dz}$ and is called the *derivative* of $f(z)$ at the point z.

Remarks: That is, $f(z)$ is differentiable at z if $\exists$ a new number $f'(z)$ associated with the point z in such a way that given an arbitrary $\epsilon > 0$ $\exists$ a $\delta(\epsilon, z)$ such that

$$\left| \frac{f(z+h) - f(z)}{h} - f'(z) \right| < \epsilon$$

whenever $z + h \in D$ and $|h| < \delta$.

Differentiability and continuity 2

Alternative Definition:

Definition

$f(z)$ is differentiable at z and has derivative $f'(z)$ there, $\iff$ for any sequence $\{z_n\}$ tending to z, the sequence $\left\{\frac{f(z_n)-f(z)}{z_n-z}\right\} \to f'(z)$.

Example

let $f(z) = z^2$. Then $f'(z) = 2z \ \forall \ z$ because

$$f'(z) = \lim_{h\to 0} \frac{(z+h)^2-z^2}{h} = \lim_{h\to 0} \frac{2zh+h^2}{h} = \lim_{h\to 0}(2z+h) = 2z$$

Remark: Differentiability$\Longrightarrow$continuity. However, the converse is not true.

Differentiability and continuity 3

Remark: Differentiability$\Longrightarrow$continuity. However, the converse is not true.

Example

The function $f(z) = |z|$ is continuous, but it is not differentiable.

Consider $$\lim_{h\to 0} \frac{|z+h|-|z|}{h} \qquad (2)$$

and let $z = r(\cos\theta + i\sin\theta)$.

For $\theta = \theta_0$, a constant, let $h = \rho(\cos\theta_0 + i\sin\theta_0)$. Then (2) becomes

$$\lim_{\rho\to 0} \frac{r+\rho-r}{\rho(\cos\theta_0+i\sin\theta_0)} = (\cos\theta_0 - i\sin\theta_0)$$

This last expression is clearly *not* independent of the approach to z.

Analytic functions 1

Definition

If a complex function $w = f(z)$ is defined and differentiable at all points of a domain D, we say that $f(z)$ is an *analytic (or regular or holomorphic)* function in D.

Remark: The statement, "$f(z)$ is an analytic function" means that $\exists$ *some* domain of analyticity.

Definition

An analytic function is said to be *regular at a point* z_0 if it is regular in a neighborhood $|z - z_0| < \epsilon$ of z_0.

Definition

z_0 is called a *singular point* of $f(z)$ if $f(z)$ is not differentiable at z_0, but if every neighborhood of z_0 contains points at which $f(z)$ is differentiable.

Remark: z^n (n a positive integer) is a regular function of z at all finite points of the complex plane.

Analytic functions 2

Remark: Given that there is a complete analogy between the definition (1) of the complex derivative and the corresponding formula

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

defining the definition of a real function, all of the standard formulae for derivatives of sums, products, quotients, etc., also hold in the complex case. Thus if $f(z)$ and $g(z)$ are differentiable at a point z and c_1 and c_2 are any complex constants, then so are $c_1 f(z) \pm c_2 g(z)$, $c_1 f(z)$, $f(z)g(z)$, $f(z)/g(z)$, (provided $g(z) \neq 0$), and

$$\begin{gathered}
[c_1 f(z) \pm c_2 g(z)] = c_1 f'(z) \pm c_2 g'(z) \\
[c_1 f(z)]' = c_1 f'(z) \\
[f(z)g(z)]' = f'(z)\,g(z) + f(z)\,g'(z) \\
\left[\frac{f(z)}{g(z)}\right]' = \frac{f'(z)g(z)-f(z)g'(z)}{g^2(z)} \qquad g(z) \neq 0
\end{gathered}$$

Analytic functions 3

Thus every *polynomial* $f(z) = a_0 + a_1 z + \cdots + a_n z^n$ (where $a_0, a_1, \ldots, a_n$ are complex constants) is regular in the finite complex plane, and every *rational* function

$$\frac{a_0 + a_1 z + \cdots + a_n z^n}{b_0 + b_1 z + \cdots b_m z^m}$$

(where the $a's$ and $b's$ are constants) is regular in the finite complex plane, except at the zeroes of the denominator.

Example

A zero of the denominator is a singular point of a rational function.

Analytic functions 4

Theorem

Suppose $w = g(z)$ is an analytic function in a domain D of the $z-$ plane, and when $z \in D$ $w = g(z)$ lies in a domain D' of the complex $w-$ plane. Also, suppose that $F[w]$ is an analytic function of w in D'. Then $f(z) = F[g(z)]$ is an analytic function of z in D.

In other words, an analytic function of an analytic function is again an analytic function.

The derivative of $f(z)$ is given by the chain rule

$$f'(z) = F'[w]\, g'(z) = F'[g(z)]\, g'(z)$$

Example: If D is $|z| < 1$, and $w = g(z) = 5z$ then D' is $|w| < 5$.
If $D : |z| > 0$ and $w = g(z) = \frac{1}{z}$, then $D' : |w < \infty|$

The Cauchy-Riemann Equations. Part 1

Suppose that $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ is differentiable at a point z. We shall show that differentiability implies a simple but characteristic property of $u(x,y)$ and $v(x,y)$.
Recall that when $f'(z)$ exists then the value of

$$f'(z) = \lim_{h\to 0} \frac{f(z+h)-f(z)}{h}$$

is independent of the way in which $h \to 0$. Let $h = \Delta x + i\Delta y$ and consider the difference quotient

$$\frac{f(z+h)-f(z)}{h} = \frac{[u(x+\Delta x,\ y+\Delta y)+iv(x+\Delta x,\ y+\Delta y)]-[u(x,y)+iv(x,y)]}{\Delta x+i\Delta y}. \tag{1}$$

The Cauchy-Riemann Equations. Part 2

We shall let $h = \Delta x + i\Delta y$ go to zero in two different ways:
1) Let $\Delta y = 0$, $\Delta x > 0$, and $\Delta x \to 0$. Then (1) becomes

$$\frac{f(z+h)-f(z)}{h} = \frac{[u(x+\Delta x,y)+iv(x+\Delta x,y)]-[u(x,y)+iv(x,y)]}{\Delta x} = \frac{u(x+\Delta x,y)-u(x,y)}{\Delta x} + i\frac{v(x+\Delta x,y)-v(x,y)}{\Delta x} \qquad (2)$$

Since the limit of the left hand side exist and equals $f'(z)$ then the limit of the right hand side must also exist, and therefore
$\lim_{\Delta x\to 0} \frac{u(x+\Delta x,y)-u(x,y)}{\Delta x}$ exists and $= Re[f'(z)]$ and $\lim_{\Delta x\to 0} \frac{v(x+\Delta x,y)-v(x,y)}{\Delta x}$ also exists and $= Im[f'(z)]$.
Hence $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$ exist at $z = x + iy$ and

$$f'(z) = u_x + iv_x \qquad (3)$$

The Cauchy-Riemann Equations. Part 3

2) Let $\Delta x = 0$, $\Delta y > 0$ and $\Delta y \to 0$. Then (1) becomes

$$\frac{f(z+h)-f(z)}{h} = \frac{u(x,y+i\Delta y)-u(x,y)}{i\Delta y} + i\frac{v(x,y+i\Delta y)-v(x,y)}{i\Delta y} \tag{4}$$

Let $\Delta y \to 0$. $\Longrightarrow$ u_y and v_y exist at the point z and

$$f'(z) = v_y - iu_y \tag{5}$$

From (3) and (5) we have that

$$u_x = v_y \qquad \text{and} \qquad u_y = -v_x \tag{6}$$

These are called the Cauchy-Riemann equations.

The Cauchy-Riemann Equations. Part 4

Theorem

A necessary condition for $f(z)$ to be differentiable at z is that the four partial derivatives with respect to x and y exist at $z = x + iy$ and are related there by Equations (6).

Theorem

If $f(z)$ is regular in a domain D, and its derivative is zero at all points of D, then $f(z)$ equals a constant in D. Equivalently, Two functions which are regular in the same domain D and whose derivatives coincide at all points of D differ in D only by a constant.

Proof.

$f'(z) = 0 \Longrightarrow u_x = u_y = v_x = v_y = 0$ throughout D. Hence $u(x, y)$ and $v(x, y)$ and thus $f(z)$ are constant throughout D.

$\square$

The Cauchy-Riemann Equations. Examples 1

Example

$f(z) = Re[z] = x$. here $u = x$ and $v = 0$ so that $u_x = 1, u_y = v_x = v_y = 0$. Thus the C-R Equations are satisfied nowhere so that $f(z) = x$ is differentiable nowhere.

Example

$f(z) = |z|^2 = x^2 + y^2$. Here $u = x^2 + y^2$ and $v = 0$, so that $u_x = 2x$, $u_y = 2y$ and $v_x = v_y = 0$. Therefore the C-R Equations hold only at the origin. Thus $z = 0$ is the only possible point where $|z|^2$ might be differentiable.

Remark: Satisfaction of the Cauchy-Riemann equations is a necessary condition for $f(z) = u + iv$ to be differentiable at a point. however, it is not sufficient as the example below shows.

The Cauchy-Riemann Equations. Examples 2

Remark: Satisfaction of the Cauchy-Riemann equations is a necessary condition for $f(z) = u + iv$ to be differentiable at a point. however, it is not sufficient as the example below shows.

Example

Let $f(z) = \sqrt{|xy|}$. The $u = \sqrt{|xy|}$ and $v = 0$. Using the definition it is not hard to show that at $(0,0)$ $u_x = u_y = 0$, whereas it is obvious that $v_x = v_y = 0$ everywhere. Hence the C-R Equations hold at the origin $z = 0$. However, $f(z)$ is not differentiable at $z = 0$ since the difference quotient

$$\frac{f(0+h)-f(0)}{h} = \frac{f(h)}{h} = \frac{\sqrt{\Delta x \Delta y}}{\Delta x + i\Delta y}$$

for $h = \Delta x + i\Delta y$. If we let $\Delta x = \alpha r$ and $\Delta y = \beta r$, where α and β are real and constant, and r is a real parameter (thus $h \to 0$ along any straight line), then

$$\lim_{h\to 0} \frac{f(h)}{h} = \lim_{\Delta x, \Delta y \to 0} \frac{\sqrt{\Delta x \Delta y}}{\Delta x + i\Delta y} = \lim_{r\to 0} \frac{r\sqrt{|\alpha\beta|}}{r(\alpha+i\beta)} = \frac{\sqrt{|\alpha\beta|}}{(\alpha+i\beta)}$$

Hence the limit is not independent of the path of approach, and therefore $f(z)$ is not differentiable at $z = 0$.

The Cauchy-Riemann Equations: necessary and sufficient conditions

Theorem

Suppose that the four partial derivatives of first order of u and v with respect to x and y exist and are continuous throughout a domain D. Then for $f(z) = u(x, y) + iv(x, y)$ to be regular in D, it is necessary and sufficient that the Cauchy-Riemann equations (6) *hold throughout D.*

Examples 1

Example

Consider the set $S = \{z \mid \ |z+4| \geq |z-i|\}$. Is S open? What are the boundary points of S?

S is not open since it contains points which are not internal points.

The boundary points of S is the set $T = \{z \mid \ |z+4| = |z-i|\} = \{z = x+iy | (x+4)^2 + y^2 = x^2 + (y-1)^2\}$ $= \{x+iy | x^2 + 8x + 16 + y^2 = x^2 + y^2 - 2y + 1\} =$ $\{x+iy | 8x + 16 = -2y + 1\} = \{x+iy | y = \frac{-15}{2} - 4x\}$

Example 2

Example

Let $K = \{ni|\ n \text{ is an integer}\}$. Prove that K is closed.

$K = \{..., -2i, -i, 0, i, 2i, ...\}$has no limit points since any deleted $\epsilon = 0.5$ neighborhood has no point in K. Thus K is vacuously closed because it contains all its limit points. There are no limit points, so K contains all of them.

Examples 3

Example

Let $W = \{1/n|$ where n is any nonzero integer$\}$. Find all limit points of W. Is W closed? Is W open?

The limit point of W is 0. W is not closed since 0 is not in W. W is not open since all its points are not internal points. In fact, none of its points are internal points.

Example 4

Example

Let S be a set of complex numbers. Prove that S is closed $\Leftrightarrow$ S contains all of its boundary points.

[$\Longrightarrow$] Let S be a closed set of complex numbers. Let z be any boundary point of S. Assume z is not an element of S. Every $\varepsilon-$neighborhood around z contains a point in S. Take $\varepsilon = \dfrac{1}{n}$, for $n = 1, 2, \ldots$ and pick a point z_n in each neighborhood. All of these points are distinct since $z_n \in S$ while z is not. $\{z_n\}$ approaches z. Therefore z is a limit point of S. Since S is closed and contains all of its limit point, $z \in S$. $\Longrightarrow\Longleftarrow$. Our original assumption that z is not in S is wrong. Therefore S contains all its limit point.

[$\Longleftarrow$] Let S be a set of complex numbers that contains all of its boundary points. Let z be a limit point of S. Then z is either a boundary point or an interior point. If it is an interior point then $z \in S$. If z is a boundary point than by the assumption $z \in S$. Thus S contains all of its limit points. S is closed.

Example 5

Example

Let z be a number in a set S of complex numbers. Prove that z is either a boundary point of S or an interior point of S.

Let z be a number in S. If z is an interior point of S we are done. If not, then there is no ε-neighborhood of z that is included in S. This means that every ε-neighborhood of z contains a point outside of S. And z is included in that ε-neighborhood. Thus z is a boundary point since every ε -neighborhood of z contains a point of S and a point not in S.

Example 6

Example

Find all limit points of the following sequences:

$a)\ z_n = 1 + (-1)^n \frac{n}{n+1}$ $\qquad$ $b)\ z_n = 1 + i^n \frac{n}{n+1}$

a)$\{\frac{1}{2}, \frac{5}{6}, \frac{1}{4}, \frac{9}{5}, \frac{1}{6}, \frac{13}{7}, ...\}$,the limit points are 0 and 2.

b)$\{1 + \frac{i}{2}, \frac{1}{3}, 1 - \frac{3}{4}i, \frac{9}{5}, 1 + \frac{5}{6}i, \frac{1}{7}, 1 - \frac{7}{8}i, \frac{17}{9}, ...\}$. The limit points are $\{0, 2, 1 + i, 1 - i\}$.

Example 7

Example

Evaluate $\lim_{z\to 2e^{\frac{\pi i}{3}}} \frac{z^3+8}{z^4+4z^2+16}$.

$$\lim_{z\to 2e^{\frac{\pi i}{3}}} \frac{z^3+8}{z^4+4z^2+16} = \lim_{z\to 2e^{\frac{\pi i}{3}}} \frac{(z+2)(z-1\pm i\sqrt{3})}{(z+1\pm i\sqrt{3})(z-1\pm i\sqrt{3})} =$$

$$\lim_{z\to 2e^{\frac{\pi i}{3}}} \frac{(z+2)}{(z+1\pm i\sqrt{3})} = \frac{(2e^{\frac{\pi i}{3}}+2)}{(2e^{\frac{\pi i}{3}}+1\pm i\sqrt{3})} =$$

$$= \frac{2(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}+\cos 0)}{2(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})+(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})} = \frac{2(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})+1}{3(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})} =$$

$$\frac{3\pm i2\sqrt{3}}{3\pm i3\sqrt{3}} = -\tfrac{1}{8}i\sqrt{3}+\tfrac{3}{8}$$

z^4+4z^2+16, roots: $\begin{matrix} 1-i\sqrt{3} \\ 1+i\sqrt{3} \\ -1+i\sqrt{3} \\ -1-i\sqrt{3} \end{matrix}$, z^3+8, roots: $\begin{matrix} -2 \\ 1-i\sqrt{3} \\ 1+i\sqrt{3} \end{matrix}$

Example 8

Example

Prove that $\lim_{z\to 0} \dfrac{\bar{z}}{z}$ does not exist.

Approach 0 along the x-axis where $y = 0$.
$\lim_{z\to 0} \frac{\bar{z}}{z} = \lim_{z\to 0} \frac{a+bi}{a-bi} = \lim_{z\to 0} \frac{a}{a} = 1.$
Approach 0 along the y-axis where $x = 0$.
$\lim_{z\to 0} \frac{\bar{z}}{z} = \lim_{z\to 0} \frac{a+bi}{a-bi} = \lim_{z\to 0} \frac{bi}{-bi} = -1.$
Since these values different $\lim_{z\to 0} \frac{\bar{z}}{z}$ does not exist.

Example 9

Example

Prove that $f(z) = z^2$ is uniformly continuous in the region $|z| < 1$.

Proof:

$$|f(z') - f(z'')| = \left|(z')^2 - (z'')^2\right| = |(z' + z'')(z' - z'')|$$
$$= |z' + z''|\,|z' - z''| \leq (|z'| + |z''|)\,|z' - z''| <$$
$2\,|z' - z''|$. Take $\delta = \dfrac{\varepsilon}{2}$. ■

Example 10

Example

Prove that $f(z) = \dfrac{1}{z}$ is not uniformly continuous in the region $|z| < 1$.

Proof: Let z_0 and $z_0 + \xi$ be any two points of the region such that $|z_0 + \xi - z_0| = |\xi| = \delta$.
Then

$$|f(z_0) - f(z_0 + \xi)| = \left| \frac{1}{z_0} - \frac{1}{z_0 + \xi} \right| = \frac{|\xi|}{|z_0|\,|z_0 + \xi|} = \frac{\delta}{|z_0|\,|z_0 + \xi|}$$

can be made larger than any positive number by choosing z_0 sufficiently close to 0. Hence the function cannot be uniformly continuous in the region. ■

Example 11

Example

Use the definition to prove that $f(z) = \bar{z}$ is not differentiable at any point.

Proof:
$\dfrac{f(z+h)-f(z)}{h} = \dfrac{\overline{z+h}-\bar{z}}{h} = \dfrac{\bar{z}+\bar{h}-\bar{z}}{h} = \dfrac{\bar{h}}{h} = \dfrac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$ for $h = \Delta x + i\Delta y$. Let $\Delta x = \alpha r$ and $\Delta y = \beta r$, where α, β, are real and constant, and r is a real parameter. Then
$\dfrac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = \dfrac{\alpha r - i\beta r}{\alpha r + i\beta r} = \dfrac{\alpha - i\beta}{\alpha + i\beta} \cdot \dfrac{\alpha - i\beta}{\alpha - i\beta} = \dfrac{(\alpha - i\beta)^2}{\alpha^2 + \beta^2}$ and
$\lim\limits_{h\to 0} \dfrac{\bar{h}}{h} = \lim\limits_{r\to 0} \dfrac{(\alpha - i\beta)^2}{\alpha^2 + \beta^2} = \dfrac{(\alpha - i\beta)^2}{\alpha^2 + \beta^2}$. Hence the limit is not independent of the path approach and therefore $f'(z)$ does not exist. ■

Example 12

Example

Use the definition to prove that $f(z) = Re\, z$ is not differentiable at any point.

Proof:

$$\frac{f(z+h)-f(z)}{h} = \frac{Re\,(z+h) - Re\,(z)}{h} = \frac{x+\Delta x - x}{\Delta x + i\Delta y} = \frac{\Delta x}{\Delta x + i\Delta y}$$

for $h = \Delta x + i\Delta y$. Let $\Delta x = \alpha r$ and $\Delta y = \beta r$, where α, β, are real and constant, and r is a real parameter. Then

$$\frac{\Delta x}{\Delta x + i\Delta y} = \frac{\alpha r}{\alpha r + \beta r} = \frac{\alpha}{\alpha + \beta} \text{ and}$$

$$\lim_{h\to 0} \frac{f(z+h)-f(z)}{h} = \lim_{r\to 0} \frac{\alpha}{\alpha+\beta} = \frac{\alpha}{\alpha+\beta}$$ which is dependent on the path of approach. ■

Example 13

Example

Show that $f(z) = |z|^2$ is differentiable at the origin, but is not regular there.

$$\frac{f(z+h)-f(z)}{h} = \frac{|z+h|^2 - |z|^2}{h} = \frac{(z+h)(\overline{z+h}) - z\overline{z}}{h} =$$

$$\frac{(z+h)(\overline{z}+\overline{h}) - z\overline{z}}{h} = \frac{z\overline{z} + z\overline{h} + \overline{z}h - z\overline{z}}{h}$$

$= \dfrac{z\overline{h} + \overline{z}h}{h} = \dfrac{2Re\left(\overline{z}h\right)}{h} = \dfrac{a\Delta x}{\Delta x + i\Delta y}$. If $z = 0$, then $a = 0$ and the quotient is 0 forcing the limit to be 0. If not, from a previous problem $\dfrac{a\Delta x}{\Delta x + i\Delta y} \to a\left(\dfrac{\alpha}{\alpha+\beta}\right)$ as $\Delta x, \Delta y \to 0$ the value depending on the path chosen. ■

Example 14. Part 1

Example

Let $f(z) = \frac{xy^2(x+iy)}{x^2+y^4} \qquad z \neq 0, \qquad f(0) = 0$

Show that $\lim_{h\to 0} \frac{f(h)-f(0)}{h} = 0$ as $h \to 0$ along any straight line.

$\dfrac{f(h)}{h} = \dfrac{\Delta x \Delta y^2(\Delta x + i\Delta y)}{(\Delta x^2 + \Delta y^4)(\Delta x + i\Delta y)} = \dfrac{\Delta x \Delta y^2}{\Delta x^2 + \Delta y^4}$ for $h = \Delta x + i\Delta y$. Let $\Delta x = \alpha r$ and $\Delta y = \beta r$, where α, β, are real and constant, and r is a real parameter. Then

$$\frac{\Delta x \Delta y^2}{\Delta x^2 + \Delta y^4} = \frac{\alpha r \beta^2 r^2}{\alpha^2 r^2 + \beta^4 r^4} = \frac{\alpha r \beta^2}{\alpha^2 + \beta^4 r^2} \to 0 \text{ as } r \to 0.$$

Example 14. Part 2

Example

Let $f(z) = \frac{xy^2(x+iy)}{x^2+y^4} \qquad z \neq 0, \qquad f(0) = 0$
Show that the limit is nonzero if $h \to 0$ along the curve $x = y^2$.

$$\frac{f(h)}{h} = \frac{\Delta x \Delta y^2 (\Delta x + i\Delta y)}{(\Delta x^2 + \Delta y^4)(\Delta x + i\Delta y)} = \frac{\Delta x \Delta y^2}{\Delta x^2 + \Delta y^4} \text{ for}$$

$h = \Delta x + i\Delta y$. Let $\Delta x = \Delta y^2$. Then we have:

$$\frac{\Delta y^4}{2\Delta y^4} = \frac{1}{2} \neq 0.$$

Thus $f'(0)$ does not exit.

Example 14. Part 3

Example

Let $f(z) = \frac{xy^2(x+iy)}{x^2+y^4} \qquad z \neq 0, \qquad f(0) = 0$

Do the Cauchy-Riemann equations hold at $z = 0$?

$$f(z) = \frac{xy^2(x+iy)}{x^2+y^4} = f(z) = \frac{x^2y^2}{x^2+y^4} + i\frac{xy^3}{x^2+y^4} \Rightarrow u = \frac{x^2y^2}{x^2+y^4}$$

and $v = \dfrac{xy^3}{x^2+y^4}$

$$u_x = \frac{(x^2+y^4)(2xy^2) - (x^2y^2)(2x)}{(x^2+y^4)^2} = \frac{2xy^6}{(x^2+y^4)^2}$$

$$v_y = \frac{(x^2+y^4)(3xy^2) - (xy^3)(4y^3)}{(x^2+y^4)^2} = \frac{3x^4y^2 - xy^6}{(x^2+y^4)^2} \neq u_x.$$

Answer: No.

Example 15. Part 1

Example

Show that $u(x,y) = e^{-x}(x\sin y - y\cos y)$ is harmonic.

$u_x = -e^{-x}(x\sin y - y\cos y) + e^{-x}\sin y$
$u_{xx} = e^{-x}(x\sin y - y\cos y) - e^{-x}\sin y - e^{-x}\sin y =$
$e^{-x}(x\sin y - 2\sin y - y\cos y)$
$v_y = e^{-x}(x\cos y - \cos y + y\sin y)$
$v_{yy} = e^{-x}(-x\sin y + \sin y + \sin y + y\cos y) =$
$e^{-x}(-x\sin y + 2\sin y + y\cos y)$
Thus $u_{xx} + v_{yy} = 0$.

Example 15. Part 2

Example

Let $u(x,y) = e^{-x}(x\sin y - y\cos y)$.
Find $v(x,y)$ such that $f(z) = u + iv$ is analytic.

$$v_y = u_x = -e^{-x}x\sin y + e^{-x}\sin y + e^{-x}y\cos y$$

$$v = \int[-e^{-x}x\sin y + e^{-x}\sin y + e^{-x}y\cos y]dy + g(x)$$

$$v = e^{-x}x\cos y - e^{-x}\cos y + e^{-x}y\sin y + e^{-x}\cos y + g(x)$$

$$v_x = e^{-x}\cos y - e^{-x}x\cos y - e^{-x}y\sin y + g'(x) =$$

$$-u_y = e^{-x}\cos y - e^{-x}x\cos y - e^{-x}y\sin y$$

$$g'(x) = 0$$

$$g(x) = C$$

$$v = e^{-x}x\cos y + e^{-x}y\sin y + C.$$

Thank you